7

Reasons Your FTAI Conception Rate Is Low

And what to do about each one — a field diagnostic guide

A low conception rate is rarely one problem. It is usually a stack of small ones. Each shaves off a few points. Together they cost you a whole calf crop.

This guide walks through the seven most common causes, in the order a good diagnosis follows: from the moment of insemination, backward to herd management.

For each cause you get four things — what happens, why it costs you pregnancies, what to check in the field, and how to fix it.

Read it once before the season. Keep it in the truck during the season.

1 Semen quality at the point of use

What happens. The semen was fine at the lab. By the time it reaches the cow, it may not be. Motility drops between the tank and the tract. You inseminate a straw that already lost its best cells and you never know.

Why it costs you. The lab report tells you what the batch was days ago, not what you are putting in the cow right now. Semen in the hands of the field is about four times more likely to fail than semen at the processing facility — roughly 8 percent farm-level rejection versus 2 percent at the lab (B.K. Stroud, 2012). Every dead straw is a guaranteed open cow.

What to check. Motility at the moment of use, not the label. Is progressive motility still where it should be after thaw? Are you checking every bull, every batch — or trusting the paperwork?

How to fix. Verify quality at the point of use. A field check on motility and progressive motility takes seconds and catches the straw that would have wasted the whole insemination. This is the exact gap MAKSA was built to close: motility and progressive motility on a smartphone, in the field, at about a dollar a test.

2 Thawing errors and cold chain

What happens. Water too hot. Water too cold. Straw left in too long or pulled too early. The tank opened too many times. Each one damages cells before the semen ever reaches the cow.

Why it costs you. Sperm cells are fragile to temperature, and a small excursion does real damage. In one documented case a swing of 1.5 °C dropped the pregnancy rate by nine points across 280 cows (Luciano Penteado). Nine points is a lot of open cows for one thawing mistake.

What to check. Is thaw water measured, not guessed? How long does the straw actually sit? How often is the tank opened in a working day? Is the neck ever exposed?

How to fix. Use a thermometer, not a thumb. Standardize thaw time. Limit tank openings. Then confirm the semen survived by checking motility at the point of use — protocol plus verification beats protocol alone.

> **Field note —** reasons 1 and 2 are the two you can catch on the spot, in the seconds before the straw goes into the cow.

3 Timing of insemination

What happens. The insemination lands too early or too late relative to ovulation. The protocol says one window; the reality on the ground drifts from it.

Why it costs you. Fertilization has a narrow window. Miss it and the best semen in the world will not make a pregnancy. In synchronized programs the timing is fixed by the protocol, so any drift in execution moves every cow off the mark at once.

What to check. Are you inseminating at the protocol-defined time, or when the crew gets to that pen? How much time passes between the first cow and the last in a large group? Is the interval from the final hormone step to insemination the one the protocol specifies?

How to fix. Hold the protocol window tight. Plan crew and pen order so the last cow is not hours behind the first. Treat the timing interval as fixed, not flexible.

4 Technician-to-technician variation

What happens. Two technicians run the same protocol on the same farm with the same semen. One gets good numbers. One does not. The difference is technique.

Why it costs you. This is one of the largest hidden variables in the field. One study found pregnancy rates varying from 15 percent to 82 percent between technicians on the same operation (Sá Filho, *Theriogenology*, 2009). That is not the semen. That is the hand.

What to check. Do you track conception rate by technician, not just by farm or by bull? Is there a gap between your best and worst performer? Is deposition site consistent, or does it wander?

How to fix. Measure per technician — you cannot fix what you do not track. Retrain the low performers on deposition and handling, and pair them with your best for a season. Small technique corrections recover large numbers of pregnancies.

15–82%

pregnancy rate spread between technicians, same operation

Sá Filho, Theriogenology, 2009

5 Protocol adherence and synchronization

What happens. A hormone step is late. A dose is skipped. An interval is shortened because the schedule got tight. The synchronization program only works if every step lands on time.

Why it costs you. These protocols are built on precise intervals. Move one step and the whole follicular wave shifts. Cows ovulate off-schedule and your fixed-time insemination no longer matches their biology.

What to check. Is every step given on the exact day and time? Are intervals between steps the ones the protocol specifies? Is anyone skipping steps to save a trip?

How to fix. Treat the protocol as non-negotiable. Build the calendar around the intervals, not around convenience. If a step cannot be given on time, that group's outcome is already compromised — plan the labor so it does not happen.

6 Body condition and nutrition

What happens. Cows go into the breeding season too thin or in negative energy balance. The reproductive system is the first thing the body shuts down when energy is short.

Why it costs you. A cow that is losing weight is a cow the body decides cannot afford a pregnancy. Low body condition score at insemination pulls conception down no matter how good your semen or timing is. Mineral gaps do the same, quietly.

What to check. What is the body condition score of the group at insemination — not two months before? Are cows gaining, holding, or losing? Is the mineral and energy program actually reaching the animals?

How to fix. Get body condition right before the season, not during it. Aim for cows gaining or holding at insemination. Close mineral and energy gaps early — nutrition is slow to fix, so start early.

7 Handling and stress at insemination

What happens. Rough handling. Long waits on the chute. Heat. A stressed cow at the moment of insemination has a body full of stress hormones that work against pregnancy.

Why it costs you. Stress hormones interfere with the signals that support conception and early pregnancy, and heat stress drops fertility further. The cow you fought onto the chute in the midday sun is not the cow you want to inseminate.

What to check. How calm is the flow through the chute? How long does a cow wait before insemination? Are you working in the heat of the day? Is the facility causing fear or injury?

How to fix. Work cattle calmly and quietly. Shorten the wait between restraint and insemination. Move the work to cooler parts of the day in hot weather. Good handling is free, and it protects every other thing you did right.

Putting it together

Walk these seven in order the next time your numbers disappoint. Most low-conception seasons are two or three of them stacked together, not one big failure.

Two of them — semen quality at the point of use, and thawing and cold chain — come down to what is actually in the cow at the moment of insemination. Those are the ones you can catch on the spot, before they cost you.

Check the straw before it becomes an open cow

MAKSA reads total and progressive motility on a smartphone, in the field, validated at r above 0.90 against laboratory CASA equipment. Over 1,000 field tests, 10-plus countries, five species.

r > 0.90	1,000+	~60s
correlation with lab CASA	field tests	per analysis, on the farm

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